Lecture 13: Comparison Tests

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- Use the direct comparison test to tell whether a series converges or diverges when possible and know the limitations of the test.
- Use the limit comparison test to tell whether a series converges or diverges when possible and know the limitations of the test.
- Identify the series to be used for the comparison test especially when the question does not suggest any.

Direct Comparison Test

Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n. Then

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Σa_n and Σb_n have non-negative terms.

$\Downarrow$

Σa_n and Σb_n converges iff their partial sum is bounded from above.

Proof:

① Suppose that Σb_n converges, then we can assume that Σb_n converges to L.

Consider the partial sum $S_N = a_1 + a_2 + \dots + a_N$

(we want to show that S_N is bounded from above)

$$S_N = a_1 + a_2 + \dots + a_N \leq b_1 + b_2 + \dots + b_N \leq \sum b_n = L$$

So S_N is bounded from above by L and hence Σa_n converges.

② Suppose that $\sum a_n$ diverges, but $\sum b_n$ converges.

Consider the partial sum

$$\sum_{n=1}^{N} b_n = b_1 + b_2 + \dots + b_N \geq a_1 + a_2 + \dots + a_N = \sum_{n=1}^{N} a_n$$

If $\sum b_n$ converges, then the partial sum $\sum_{n=1}^{N} b_n$ is bounded from above by a number M.

In this case, $\sum_{n=1}^{N} a_n$ must also be bounded from above by M.

However, this implies that $\sum a_n$ converges, a contradiction.

So, $\sum b_n$ must be divergent.

Direct Comparison Test

Example: The series $\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges.

Note that $\frac{5}{5n-1} \geq \frac{5}{5n} = \frac{1}{n} > 0$ for all n

$\sum \frac{1}{n}$ is a divergent harmonic series.

So $\sum \frac{5}{5n-1}$ diverges by Direct Comparison Test.

Direct Comparison Test

Example: The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

$$= 1 + 1 + \frac{1}{2\times 1} + \frac{1}{3\times 2\times 1} + \frac{1}{4\times 3\times 2\times 1} + \cdots$$

$$< 1 + 1 + \frac{1}{2} + \frac{1}{2\times 2} + \frac{1}{2\times 2\times 2} + \cdots$$

$$= 1 + \underbrace{1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots}_{\text{converges}}$$

Geometric series with $a=1$, $r=\frac{1}{2}$, $|r|<1$ (converges)

So, by DCT, $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

Direct Comparison Test

Example: The following series converges.

$$5+\frac{2}{3}+\frac{1}{7}+1+\frac{1}{2+\sqrt{1}}+\frac{1}{4+\sqrt{2}}+\frac{1}{8+\sqrt{3}}+\cdots+\frac{1}{2^n+\sqrt{n}}+\cdots$$

$$\sum_{n=0}^{\infty}\frac{1}{2^n+\sqrt{n}}$$

Note that $\frac{1}{2^n+\sqrt{n}} \leq \frac{1}{2^n}$ for all $n \geq 0$

$\sum_{n=0}^{\infty}\frac{1}{2^n}$ is a geometric series with $a=1$, $r=\frac{1}{2}$ (converges)

By DCT, $\sum_{n=0}^{\infty}\frac{1}{2^n+\sqrt{n}}$ converges. Since finite terms to a convergent series does not affect its convergent behaviour, so $5+\frac{2}{3}+\frac{1}{7}+\sum_{n=3}^{\infty}\frac{1}{2^n+\sqrt{n}}$ converges.

Direct Comparison Test

Example: Does $\sum_{n=1}^{\infty} \frac{n+1}{n^2}$ converge or diverge?

Note that $\frac{n+1}{n^2} = \frac{n}{n^2} + \frac{1}{n^2}$

$= \frac{1}{n} + \frac{1}{n^2}$

$\geq \frac{1}{n} > 0$

Again $\sum \frac{1}{n}$ is a divergent harmonic series.

By Direct Comparison Test, $\sum \frac{n+1}{n^2}$ also diverges.

Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$, where N is an integer.

1. If $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Proof: (1) $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = c > 0$, by the definition of convergence,

$\forall \varepsilon > 0$, there exists an integer N such that

$$\left| \frac{a_n}{b_n} - c \right| < \varepsilon \quad \text{whenever} \quad n > N.$$

$c > 0 \Rightarrow \frac{c}{2} > 0$, so we can take $\varepsilon = \frac{c}{2}$

$$\Rightarrow \left| \frac{a_n}{b_n} - c \right| < \frac{c}{2}$$

$$\Rightarrow -\frac{c}{2} < \frac{a_n}{b_n} - c < \frac{c}{2}$$

$$\Rightarrow 0 < \frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2}$$

$$\Rightarrow 0 < \frac{c}{2} b_n < a_n < \frac{3c}{2} b_n \quad (b_n > 0)$$

If $\sum b_n$ converges, then $\left(\frac{3c}{2}\right) b_n$ converges.

By DCT, $\sum a_n$ converges.

If $\sum b_n$ diverges, then $\left(\frac{c}{2}\right) b_n$ diverges.

By DCT, $\sum a_n$ diverges.

(2) $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$

for all $\varepsilon > 0$, there exists an integer N such that for all $n > N$,

$$\left| \frac{a_n}{b_n} - 0 \right| < \varepsilon$$

Take $\varepsilon = 1$, $\left| \frac{a_n}{b_n} - 0 \right| < 1$

$$-1 < \frac{a_n}{b_n} < 1$$

$$-b_n < a_n < b_n$$

Since $a_n > 0$, $0 < a_n < b_n$

By Direct Comparison Test,

if $\sum b_n$ converges, then $\sum a_n$ converges.

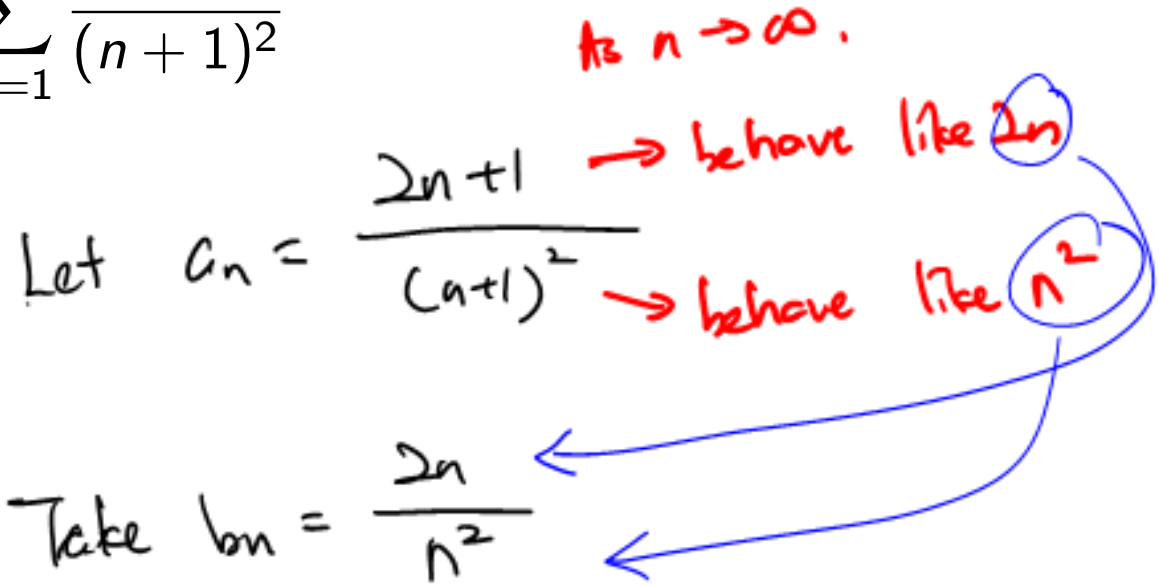
(3) $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$. Then there exists an integer N such that for all $n \geq N$, $\frac{a_n}{b_n} > 1$ ($\varepsilon = 1$) $\Rightarrow a_n > b_n > 0$

If $\sum b_n$ diverges, by DCT,

$\sum a_n$ diverges.

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2}$



$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2n+1}{(n+1)^2}}{\frac{2}{n}} = \lim_{n\to\infty} \frac{2n^2+n}{2n^2+4n+2}$$

$$= \lim_{n\to\infty} \frac{2+\frac{1}{n}}{2+\frac{4}{n}+\frac{2}{n^2}} = 1 > 0$$

(case 1)

$$\sum \frac{2}{n} = 2\sum \frac{1}{n} \quad \text{(diverges)}$$

$\sum \frac{1}{n}$ ← divergent harmonic series.

So by Limit Comparison Test,

$$\sum \frac{2n+1}{(n+1)^2} \quad \text{diverges.}$$

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

As $n \to \infty$ behave like 1

Take $a_n = \frac{1}{2^n - 1}$ → behave like 2^n

Take $b_n = \frac{1}{2^n}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{2^n - 1}}{1/2^n} = 1 \quad \text{(case 1)}$$

$\sum \frac{1}{2^n}$ is a geometric series with $a = \frac{1}{2}$, $r = \frac{1}{2}$ (converges)

So by LCT, $\sum \frac{1}{2^n - 1}$ also converges.

Limit Comparison Test

Example: $\sum_{n=2}^{\infty} \frac{1 + n \ln n}{n^2 + 5}$

As $n \to \infty$

Let $a_n = \frac{1 + n \ln n}{n^2 + 5}$ → behave like $n \ln n$

→ behave like n^2

Take $b_n = \frac{n \ln n}{n^2} = \frac{\ln n}{n}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} \lim_{n\to\infty} \frac{\frac{1 + n \ln n}{n^2 + 5}}{\frac{\ln n}{n}} = \lim_{n\to\infty} \frac{n + n^2 \ln n}{n^2 \ln n + 5 \ln n}$$

$$= \lim_{n\to\infty} \frac{\frac{n}{n^2 \ln n} \to 0 + 1}{1 + \frac{5}{n^2} \to 0} = 1 > 0$$

Note that $\frac{\ln n}{n} \geq \frac{1}{n} \geq 0$ for $n \geq 3$

$(\ln n \geq 1 \text{ for } n \geq e^1)$

and $\sum_{n=3}^{\infty} \frac{1}{n}$ is a divergent harmonic series.

So, by DCT, $\sum_{n=3}^{\infty} \frac{\ln n}{n}$ diverges and hence $\sum_{n=2}^{\infty} \frac{\ln n}{n}$ diverges.

By Limit Comparison Test, $\sum_{n=2}^{\infty} \frac{1+n \ln n}{n^2+5}$ also diverges.

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{\ln n}{n^{\frac{3}{2}}}$

As $n \to \infty$

Let $a_n = \frac{\ln n}{n^{3/2}}$ → $\ln n$, $n^{3/2}$, $n^{5/4}$

$= \frac{\ln n}{n^{5/4+1/4}} = \frac{\ln n}{n^{5/4} n^{1/4}}$ → $\ln n$, $n^{5/4}$ → grows a lot faster than $\ln n$

Take $b_n = \frac{1}{n^{5/4}}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\frac{\ln n}{n^{3/4} \cdot n^{1/4}}}{1/n^{3/4}} = \frac{\ln n}{n^{1/4}} = ?$$

Let $f(x) = \dfrac{\ln x}{x^{1/4}}$ cts at $[1, \infty)$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\ln x}{x^{1/4}}$$

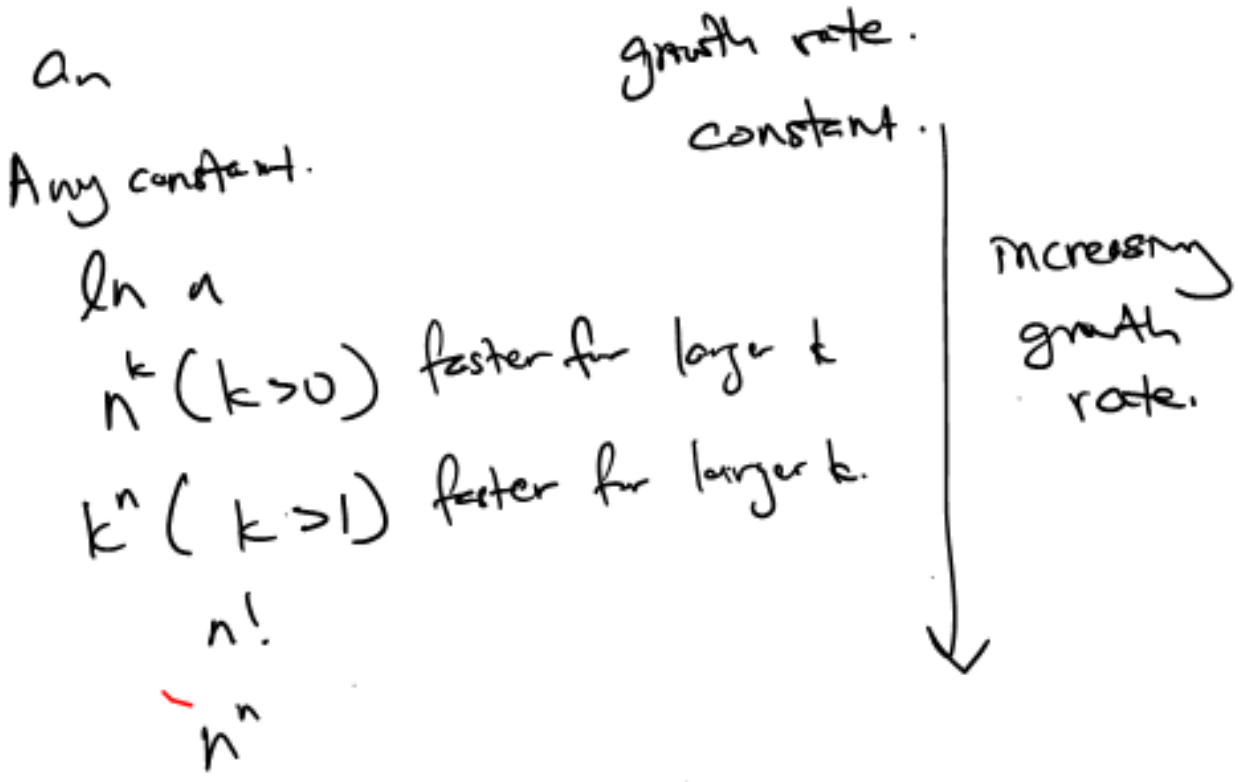
$$= \lim_{x\to\infty} \frac{1/x}{\frac{1}{4} x^{-3/4}} \quad \text{L'Hôpital rule.}$$

$$= \lim_{x\to\infty} \frac{4}{x^{1/4}} = 0 \quad \text{(case 2)}$$

$\sum \dfrac{1}{n^{3/4}}$ is a p-series with $p = 3/4 > 1$ (convergent)

So, by Limit Comparison Test, $\sum a_n$ also converges.

Extra (growing rate of sequence)



This table provides you some "guesses" of whether a sequence converges / diverges. However, you need to verify this with a proper method.

Eg: $a_n = \frac{n!}{n^n}$ $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n!}{n^n} = 0$

$n!$ → grow very fast

n^n → grow even faster

But to prove this, we can use the Sandwich Theorem.

Limit Comparison Test

Example: $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)}$

As $n \to \infty$

Let $a_n = \frac{1}{n(\ln n)} \to 1$

$\to n$ (n grow faster than $\ln n$)

Take $b_n = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{n \ln n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{1}{\ln n} = 0 \quad (\text{case 2})$$

But, $\sum b_n = \sum \frac{1}{n}$ diverges. In this case, we cannot use LCT. We need to use other method, for example Integral Test (Exercise).